

Math 285 – James Scholar Project
Predator-Prey Models: Numerical Simulation

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Task 1: Equilibria Analysis

System (1.1) – Classic Lotka-Volterra

The system is:

$$\begin{aligned}p' &= -dp + apy \\ y' &= ky - bpy\end{aligned}$$

Setting $p' = 0$ and $y' = 0$:

$$\begin{aligned}p(-d + ay) &= 0 \quad \rightarrow \quad p = 0 \quad \text{or} \quad y = d/a \\ y(k - bp) &= 0 \quad \rightarrow \quad y = 0 \quad \text{or} \quad p = k/b\end{aligned}$$

This gives two non-negative equilibria:

- Extinction equilibrium: $(p^*, y^*) = (0, 0)$. Both populations die out.
- Coexistence equilibrium: $(p^*, y^*) = (k/b, d/a)$. Both species persist.

How each parameter affects the equilibrium:

- d (predator death rate): raises $y^* = d/a$. A higher death rate requires more prey to sustain the predator population.
- a (predator benefit rate): lowers $y^* = d/a$. More efficient predators need fewer prey at equilibrium.
- k (prey growth rate): raises $p^* = k/b$. Faster-reproducing prey can support more predators.
- b (predation rate): lowers $p^* = k/b$. Heavier predation keeps the equilibrium predator count lower — a counterintuitive result known as the Volterra principle.

The number of equilibria is always exactly 2 for any positive parameter values.

System (1.2) – With Prey Carrying Capacity M

The prey equation becomes $y' = ky(M - y) - bpy$. Setting $p = 0$ gives $y = 0$ or $y = M$. Setting $y = d/a$ (from $p' = 0$, $p \neq 0$):

$$k(M - d/a) = bp \quad \rightarrow \quad p^* = k(aM - d) / (ab)$$

Equilibria:

- $(0, 0)$ – both extinct.
- $(0, M)$ – prey at carrying capacity, no predators. This equilibrium does not exist in System (1.1).
- $(k(aM - d)/(ab), d/a)$ – coexistence, exists only when $aM > d$.

The new $(0, M)$ equilibrium is biologically important: if predators die out, prey stabilise at their carrying capacity rather than growing without bound.

System (1.3) – With Bounded Consumption Rate

Interaction terms become $py/(1+y)$, capping each predator's consumption rate. Setting $p = 0$ is unchanged, giving $(0, 0)$ and $(0, M)$. For $p \neq 0$:

$$ay/(1+y) = d \rightarrow y^* = d/(a-d) \quad [\text{requires } a > d]$$

Substituting back:

$$p^* = ak(M - d/(a-d)) / (a-d)$$

If $a \leq d$, no coexistence equilibrium exists at all — predators cannot survive even with unlimited prey because consumption is bounded below their death rate. This constraint is absent from the earlier models.

Task 2: Numerical Simulation (Diving Deeper Option 2)

Method: Runge-Kutta Order 4 (RK4)

RK4 was implemented in Python for the 2D system $p' = f1(p,y)$, $y' = f2(p,y)$. At each time step h , four slope estimates are computed for each variable:

$$\begin{aligned} k1p &= f1(p_n, y_n) \\ k2p &= f1(p_n + h/2*k1p, y_n + h/2*k1y) \\ k3p &= f1(p_n + h/2*k2p, y_n + h/2*k2y) \\ k4p &= f1(p_n + h*k3p, y_n + h*k3y) \\ p_{n+1} &= p_n + (h/6)(k1p + 2*k2p + 2*k3p + k4p) \end{aligned}$$

The same four slopes are computed in parallel for y using $f2$. A step size of $h = 0.005$ was used. The full Python code is included as a separate file with this submission.

System (1.1): Various Initial Conditions

Parameters: $d = 0.5$, $a = 0.02$, $k = 1.0$, $b = 0.04$. Coexistence equilibrium: $(p^*, y^*) = (25, 25)$. Four initial conditions were run from $t = 0$ to $t = 80$.

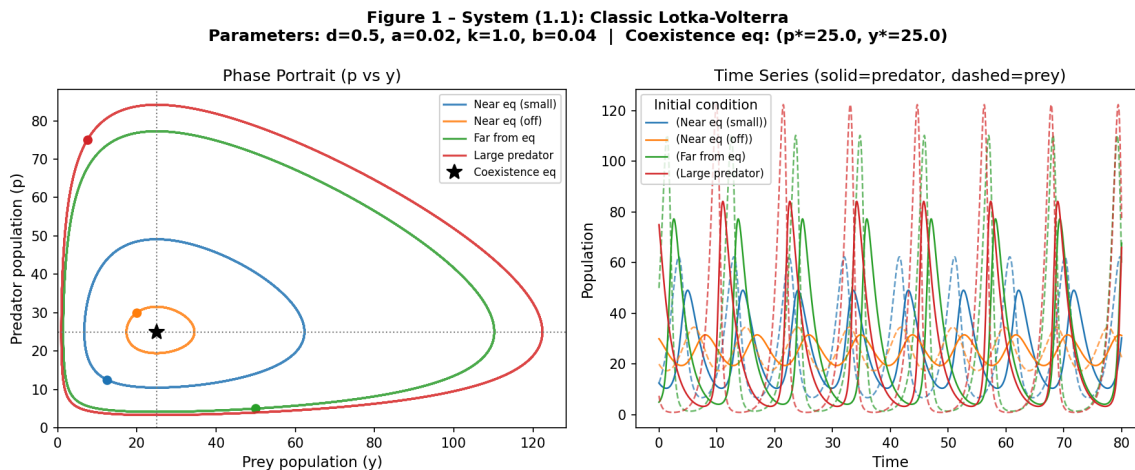


Figure 1. Phase portrait (left) and time series (right) for System (1.1) with four initial conditions.

Every initial condition produces a closed loop in the phase portrait, orbiting the coexistence equilibrium indefinitely. This is the defining feature of the classic Lotka-Volterra model: solutions neither converge to nor diverge from the equilibrium, but cycle around it forever. Biologically, when prey are plentiful, predators thrive and grow; increased predation drives prey numbers down; fewer prey cause predator decline; prey then recover; and the cycle repeats. Initial conditions far from the equilibrium produce wider oscillations, which in nature could bring a population close to local extinction before recovery.

Effect of Parameter k (Prey Growth Rate)

Figure 2 - Effect of varying k (prey growth rate) - System (1.1)
Fixed: $d=0.5$, $a=0.02$, $b=0.04$; Initial condition: $(p=5, y=10)$

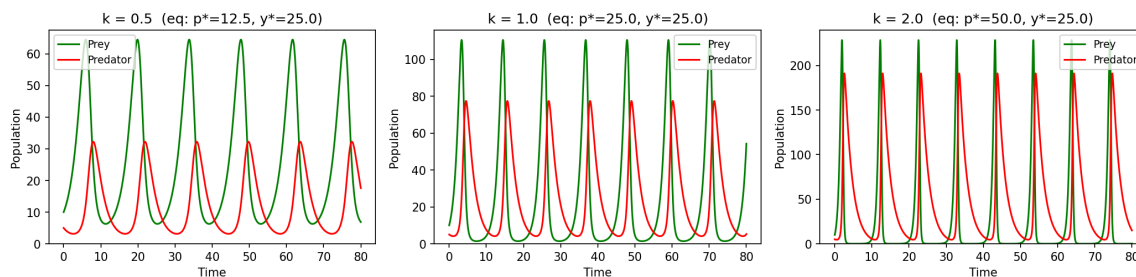


Figure 2. Time series for System (1.1) with $k = 0.5, 1.0$, and 2.0 (all other parameters fixed). IC = $(p=5, y=10)$.

As k increases, the oscillation period shortens — populations cycle faster. Faster-reproducing prey recover more quickly after being eaten, shortening each boom-bust cycle. The equilibrium predator count $p^* = k/b$ also rises with k , since higher prey productivity supports a larger predator population at equilibrium.

Effect of Parameter d (Predator Death Rate)

Figure 3 - Effect of varying d (predator death rate) - System (1.1)
Fixed: $a=0.02$, $k=1.0$, $b=0.04$; Initial condition: $(p=5, y=10)$

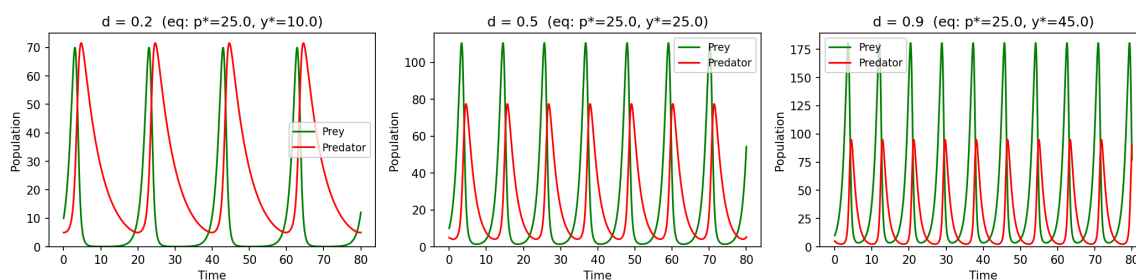


Figure 3. Time series for System (1.1) with $d = 0.2, 0.5$, and 0.9 . IC = $(p=5, y=10)$.

Higher predator death rate d raises the prey equilibrium $y^* = d/a$ while $p^* = k/b$ stays the same. In the simulations, high d produces large prey oscillations: the predator population periodically crashes, giving prey a window to overshoot dramatically before predators recover. With low d , predators persist more steadily and keep prey numbers in check.

System (1.3): Effect of Carrying Capacity M

Parameters: $d = 0.5$, $a = 2.0$, $k = 0.5$, $b = 1.0$. Three values of M were tested: small ($M = 5$), moderate ($M = 10$), and large ($M = 50$).

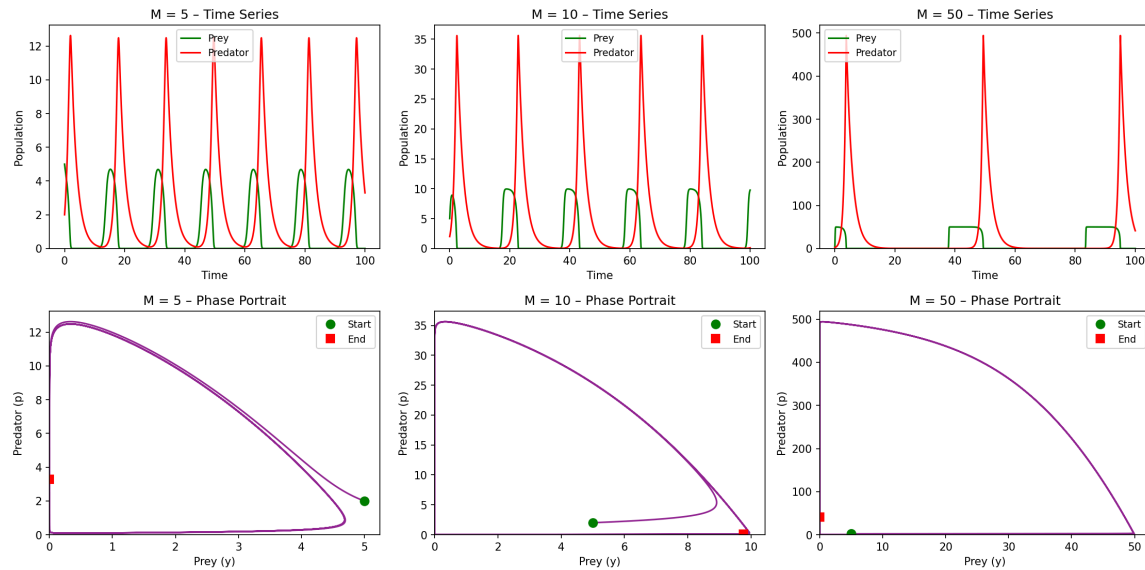


Figure 4. System (1.3) with $M = 5, 10$, and 50 . Top row: time series. Bottom row: phase portraits. IC = $(p=2, y=5)$.

Small M ($M = 5$): The phase portrait spirals inward to a fixed point — a stable coexistence equilibrium. This is a major qualitative change from System (1.1): oscillations are damped and both species settle to constant values.

Moderate M ($M = 10$): The system still converges to a stable equilibrium, but the transient oscillations are more pronounced before settling.

Large M ($M = 50$): With an effectively unconstrained prey supply, the system returns to oscillatory behaviour resembling System (1.1). A large M essentially removes the stabilising effect of the carrying capacity.

Summary

The three systems show a clear progression from idealised to realistic predator-prey dynamics. System (1.1) produces perpetual, undamped oscillations. Adding a carrying capacity and bounded consumption rate (System 1.3) stabilises the system into a true fixed-point coexistence when M is small, but recovers oscillatory behaviour for large M .

A surprising result: in System (1.1), the predator death rate d does not affect the equilibrium predator count p^* at all — only the prey equilibrium y^* changes. This is the Volterra principle, a classic counterintuitive result in mathematical ecology.

Overall, the RK4 simulations confirm the analytical equilibria from Task 1 and show how small changes in model assumptions can fundamentally shift whether species coexist stably, oscillate indefinitely, or go extinct.